

DESIGN DOCUMENT FOR HW-4

GROUP-31

Q-1 : Brief overview of your implementation.

Event based implementation

Key Changes made:

1. Multiple clients support

- Added an array `client_socks[MAX_CLIENTS]` to store all connected client sockets.

2. Modified select loop

- `select()` now monitors:
 - `STDIN` → server input
 - listener socket → new connections
 - All `client_socks[i]` → incoming messages from any client

3. Accept multiple clients

- When the listener is ready, `accept()` is called repeatedly to accept new connections.
- New sockets stored in `client_socks[]`.

4. Server messages

- Server messages sent to the first active client

```
saksham@SAKSHAMS-PC:~/CS$ ./server
Server listening on port 5432...
New client connected: fd 4
New client connected: fd 5
New client connected: fd 6
New client connected: fd 7
HI THIS IS THE SERVER RUNNING WITH 4 CLIENTS
Client 4 says: CLIENT 1

Client 5 says: CLIENT 2

Client 6 says: CLIENT 3

Client 7 says: CLIENT 4
```

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CS
saksham@SAKSHAMS-PC:~/CS$ ./client localhost
No activity for 30 secs.
HI THIS IS THE SERVER RUNNING WITH 4 CLIENTS
```

Thread Based implementation

Key Changes :

1. Thread per client
 - Each client connection spawns a new thread (pthread_create) that handles receiving messages from that client only.
2. Server input handled in separate thread
 - A dedicated server_input_thread reads from STDIN continuously.
 - Sends server messages to one random active client.

3. Mutex for client sockets

- Protects `client_socks[MAX_CLIENTS]` from concurrent access by multiple threads.

4. Concurrency

- Each client thread runs independently; multiple clients can send messages simultaneously.
- Server can send messages at the same time via its input thread.

5. Client disconnect handling

- Each client thread detects disconnection (`recv() ≤ 0`), closes socket, removes it from the array.

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CS
saksham@SAKSHAMS-PC:~/CS$ make
gcc -Wall -Werror -O3 -o server server.c -lpthread
saksham@SAKSHAMS-PC:~/CS$ ./server
Threaded server listening on port 5432...
New client connected: fd 4
New client connected: fd 5
New client connected: fd 6
New client connected: fd 7
HI I AM THE SERVER
Client 4 says: hi i am client 4

Client 5 says: hi i am client 5

Client 6 says: hi i am client 6

Client 7 says: hi i am client 7
```

```
saksham@SAKSHAMS-PC:/mnt/c/windows/system32$ cd ~/CS
saksham@SAKSHAMS-PC:~/CS$ ./client localhost
No activity for 30 secs.
HI I AM THE SERVER

hi i am client 4
No activity for 30 secs.
```

2. Are messages sent by all four clients displayed on the server's terminal?

Yes as shown by the snapshots above and explained in the implementation as well all of the clients outputs are displayed

3. Name and Roll Numbers

Dhruv Jain - 2023199

Shivank Rajput - 2023508

Altamash Hasnain - 2023073

Divyanshi - 2023209

Saksham Arora - 2023466